

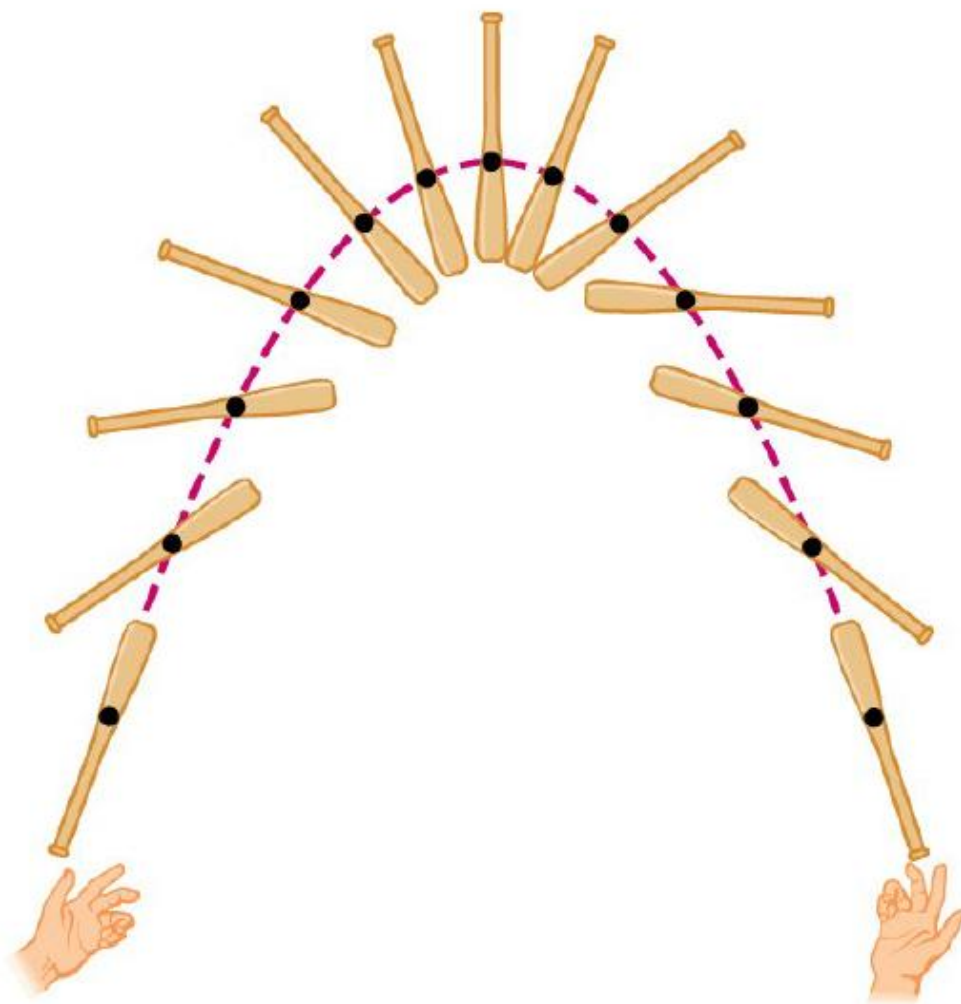
SYSTEMS OF PARTICLES

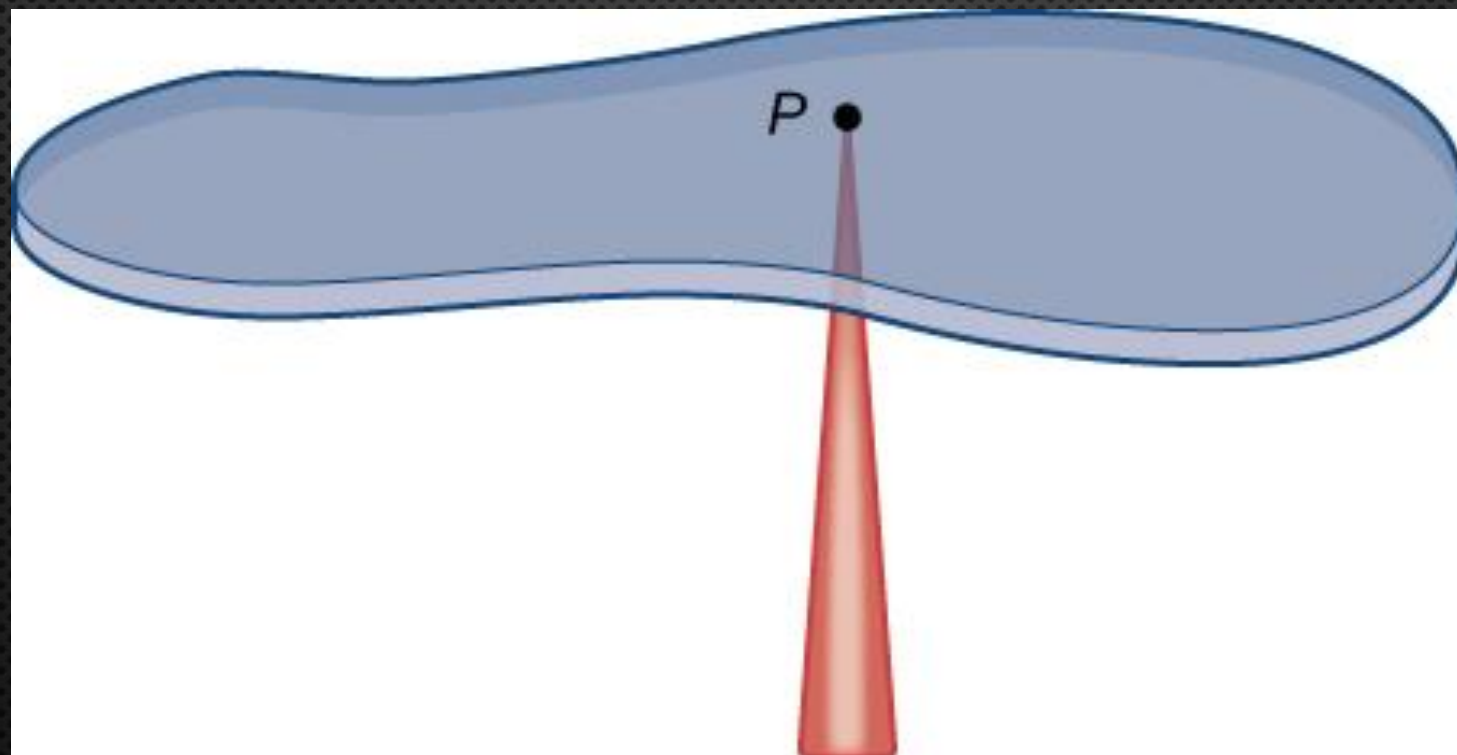
CHAPTER 9



LIST OF TERMS

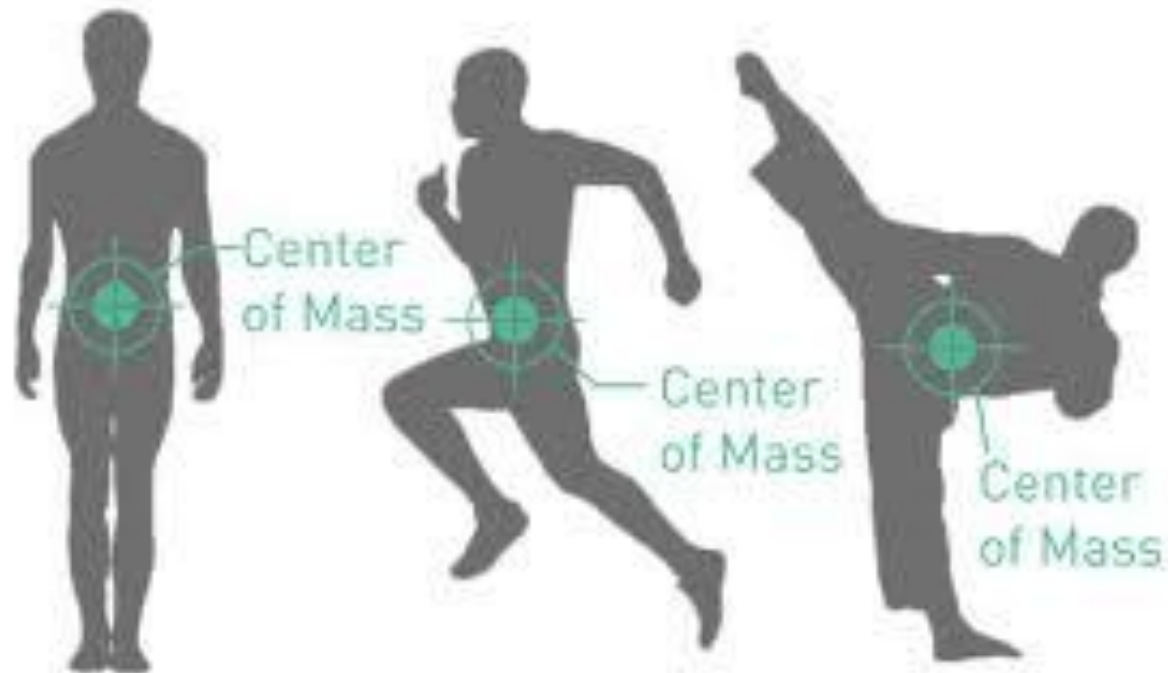
- DEFORMABLE SYSTEM
- RIGID SYSTEM
- INTERNAL FORCES
- EXTERNAL FORCES
- CENTER OF MASS
- CENTER OF GRAVITY
- THEOREM OF THE CENTER OF MASS





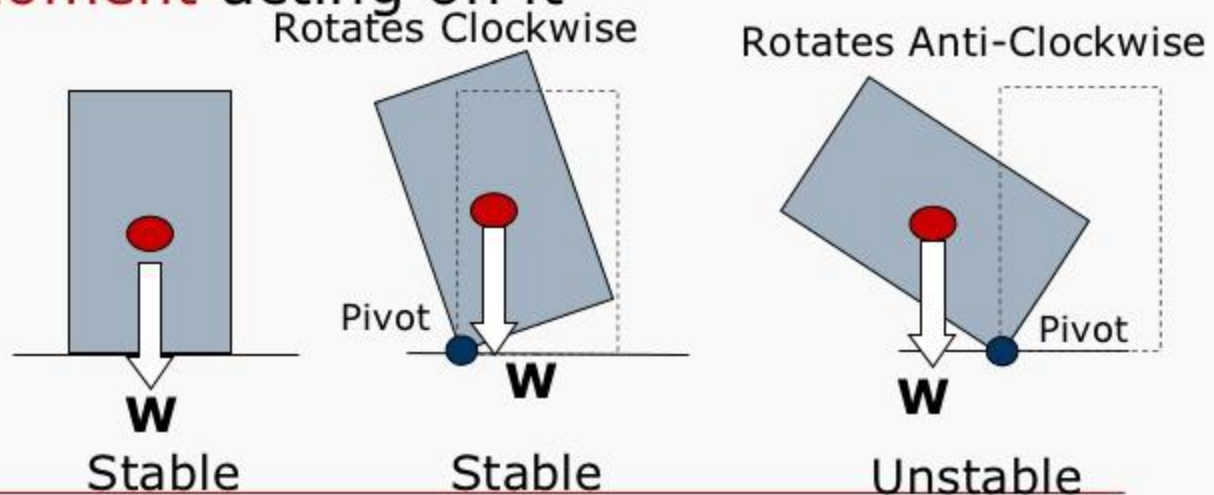






Stability

- For an object to start rotating it needs to have an **unbalanced moment** acting on it-



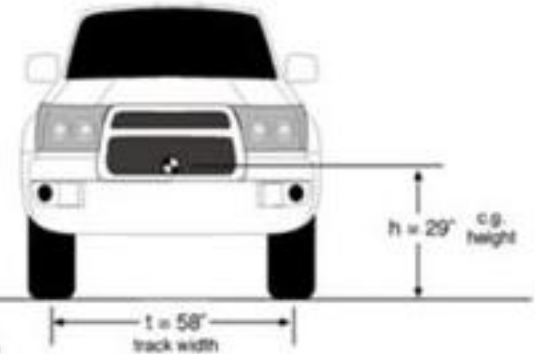
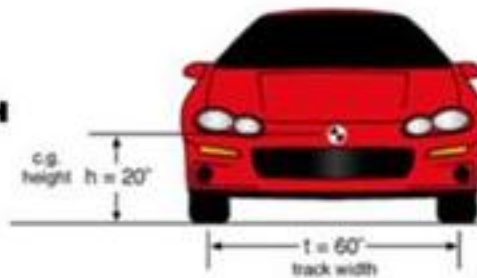


Center of Gravity Height, H

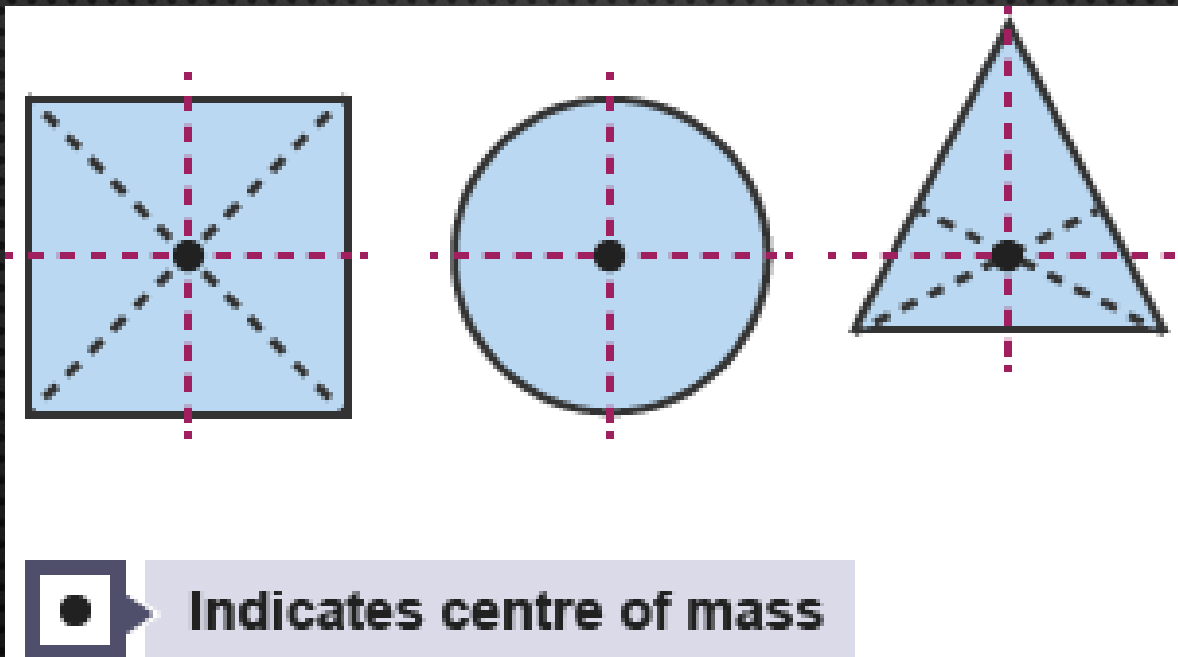
Rollover Ratings Static Stability Factor

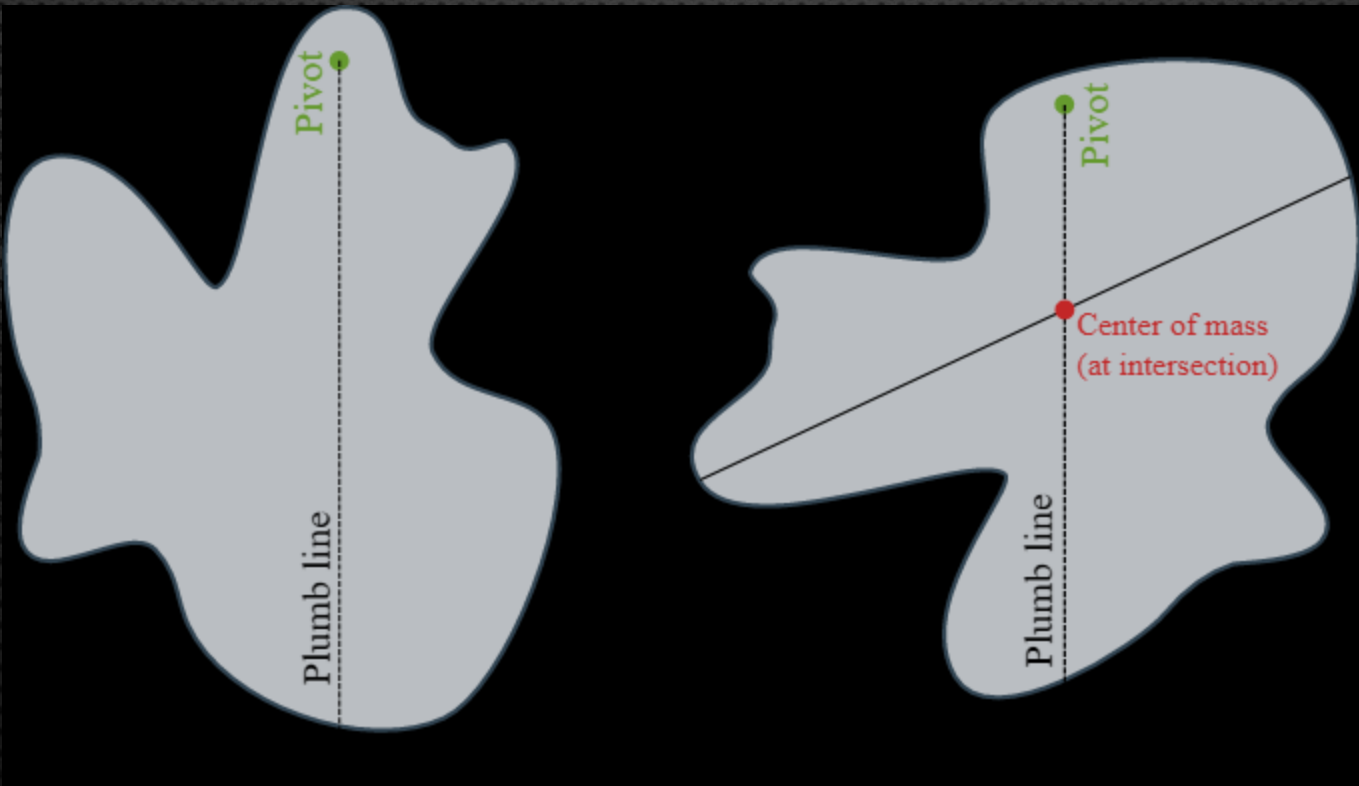
1.50

1.00



$$SSF = \frac{t}{2h}$$





PARTICLES AND SYSTEMS

- WHEN THE DISTANCE TRAVELED BY AN OBJECT IS VERY LARGE COMPARED TO ITS DIMENSIONS, THIS OBJECT CAN BE CONSIDERED AS A **PARTICLE**.
- A GROUP OF PARTICLES IS CALLED **SYSTEM OF PARTICLES**.

DEFORMABLE SYSTEM VS. RIGID SYSTEM

System



Deformable: the distance between its points varies.



Non-deformable: the distance between any two points remains fixed.

MASS CENTER OF A PARTICLES SYSTEM - 2 DIMENSIONS

- SUPPOSE A SYSTEM (S) CONSISTS OF SEVERAL PARTICLES, $P_1, P_2, P_3, \dots, P_n$, OF RESPECTIVE MASSES $m_1, m_2, m_3, \dots, m_n$.
- THEN THE POSITION OF THE CENTER OF MASS " G " OF THE SYSTEM WITH RESPECT TO AN ORIGIN " O " IS GIVEN BY :

$$x_G = \frac{m_1 \cdot x_1 + m_2 \cdot x_2 + m_3 \cdot x_3 + \dots + m_n \cdot x_n}{m_1 + m_2 + m_3 + \dots + m_n}$$

$$y_G = \frac{m_1 \cdot y_1 + m_2 \cdot y_2 + m_3 \cdot y_3 + \dots + m_n \cdot y_n}{m_1 + m_2 + m_3 + \dots + m_n}$$

$$x_G = \frac{1}{M} \sum_{i=1}^{i=n} m_i \cdot x_i$$

$$y_G = \frac{1}{M} \sum_{i=1}^{i=n} m_i \cdot y_i$$

Where: $M = m_1 + m_2 + m_3 + \dots + m_n$

1 Tell whether the following systems are rigid or deformable:

- a- A sphere rolling on an inclined plane.
- b- A flying bird.
- c- A hollow sphere, swinging under the action of its weight.
- d- An amount of milk in a shaking bottle.

9 Three particles of masses 10 g, 20 g, and 30 g are placed on one axis as in figure 9.18. Determine the center of mass of the system thus formed.

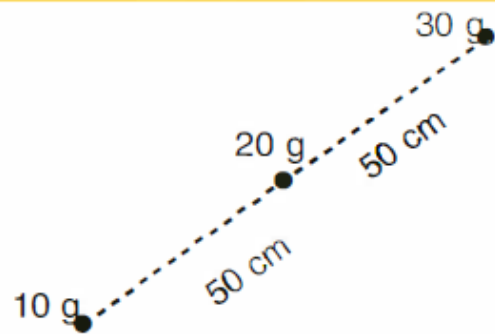
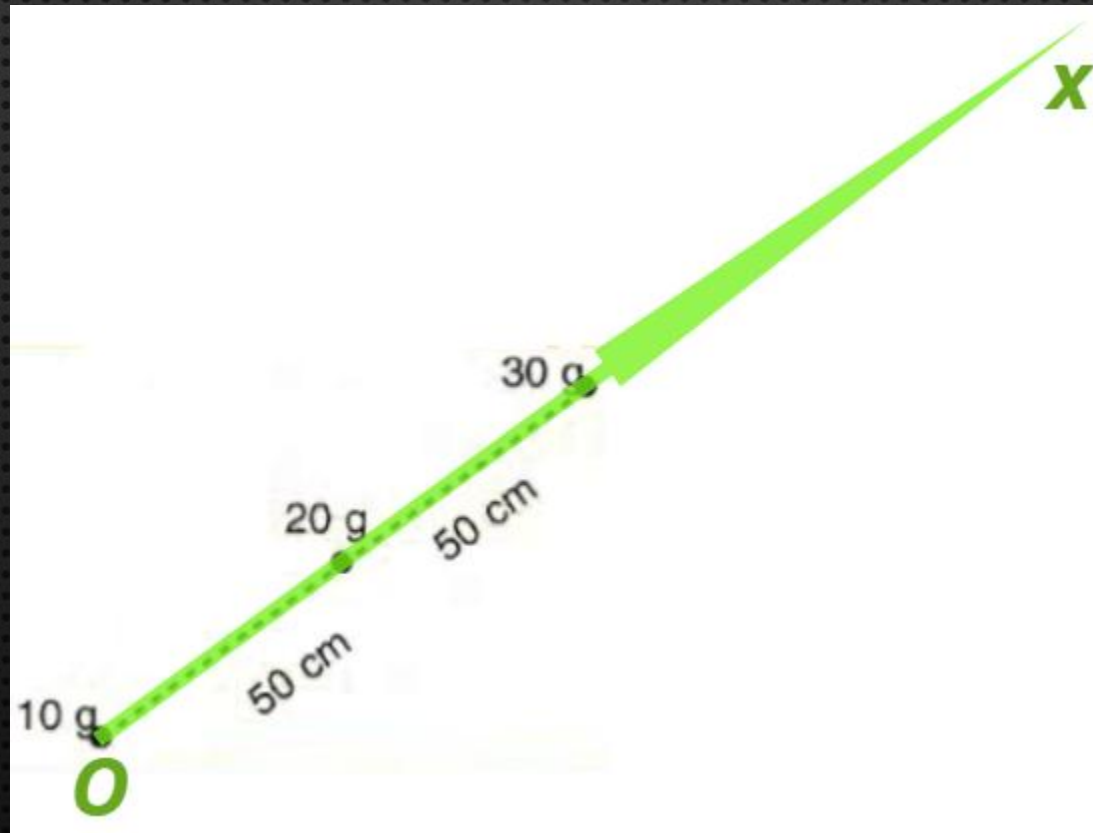
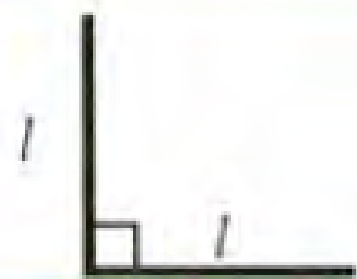


Figure 9.18 - Position of the center of mass.



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Determine the center of mass of a uniform homogeneous wire bent as in each of the cases shown in figure 9.20.



(a)

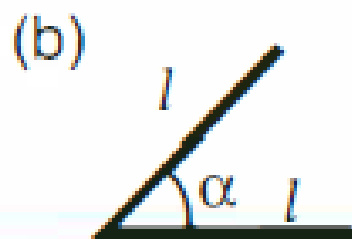
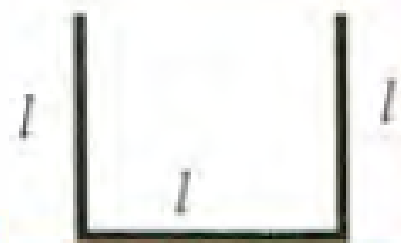
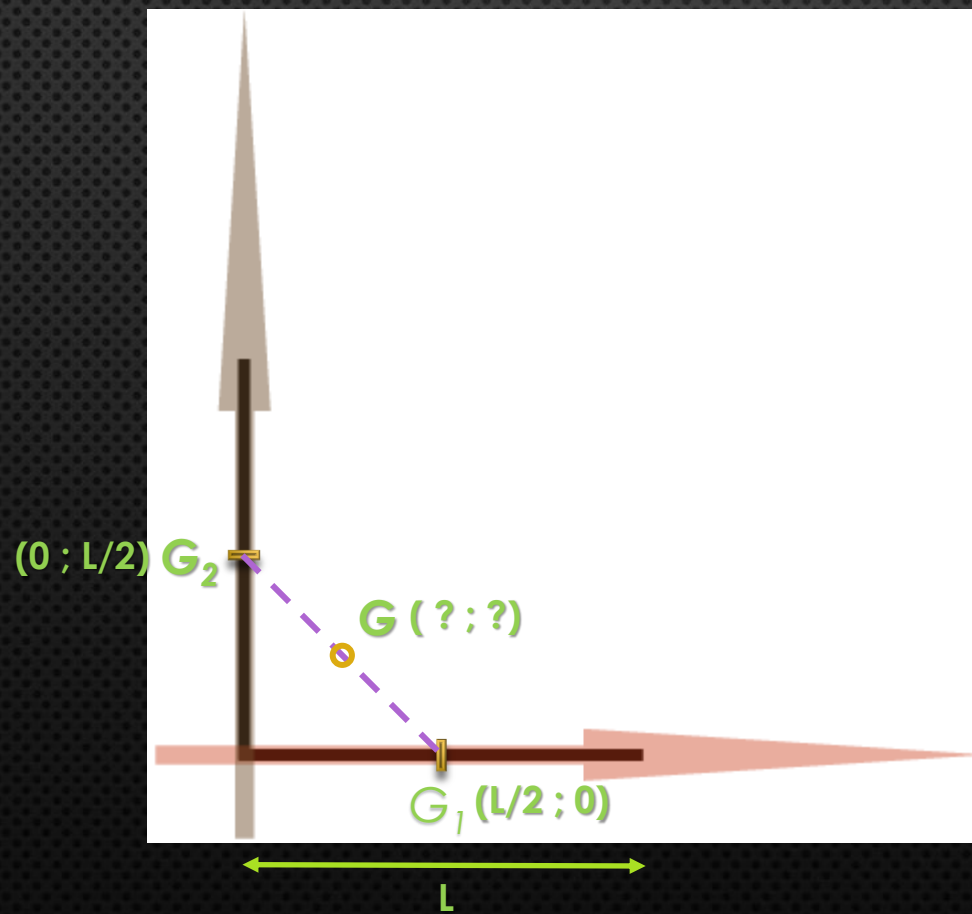


Figure 9.20

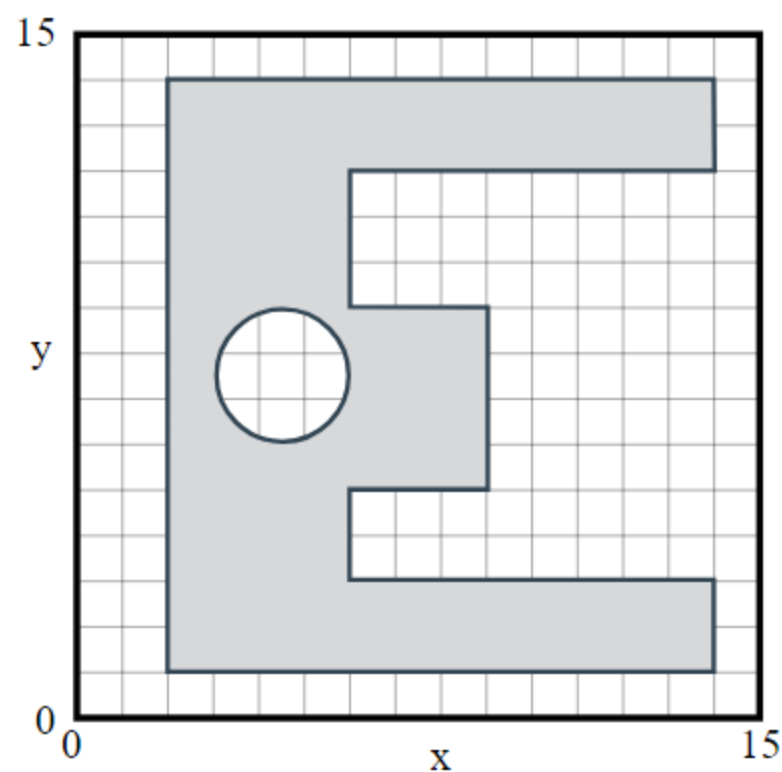


(c)

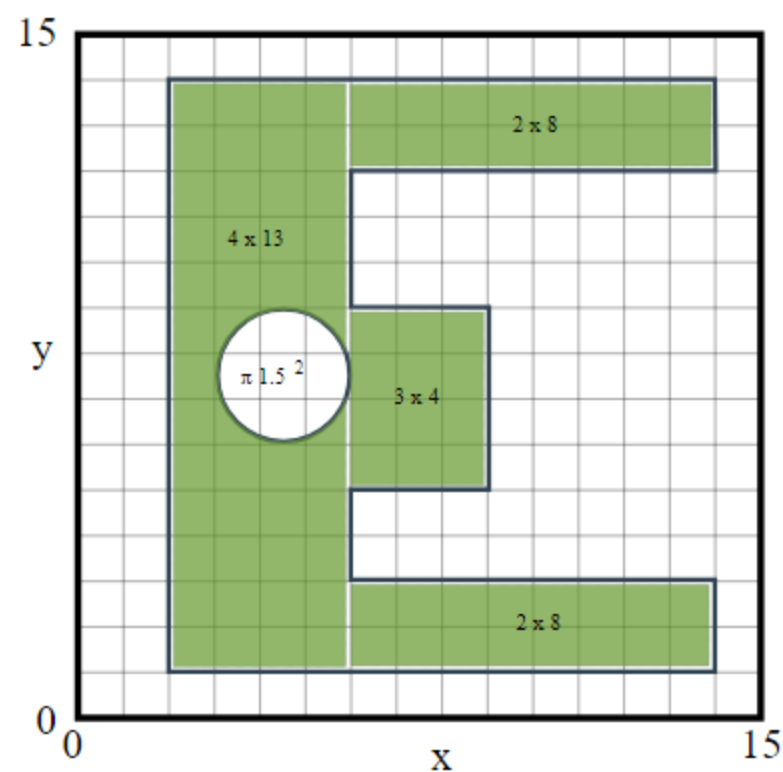


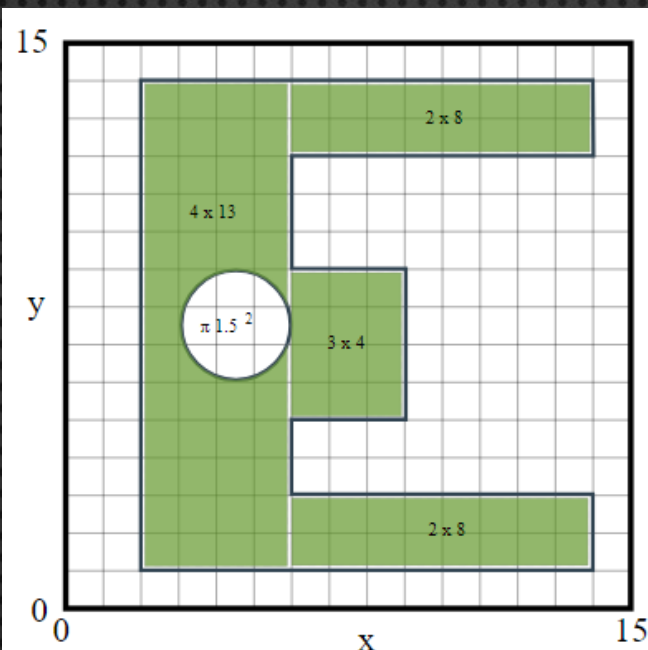
Consider the irregularly-shaped flat, uniform density object shown in Figure 3a.

(a)



(b)



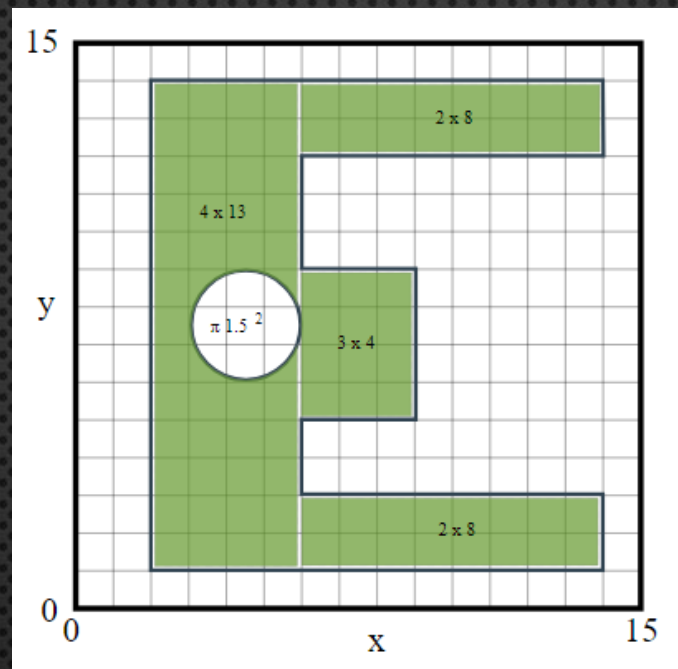


We can break this object up into four rectangles and one circle as shown in figure 3b. Here we are only interested in the position of the center of mass in the relative units shown in the figure. The material has uniform density so the mass is proportional to the area. For simplicity we can represent the mass of each section in units of 'squares' as shown in the diagram.

In the x direction, the center of mass is at:

$$\frac{16 \cdot 10 + 52 \cdot 4 + 12 \cdot 7.5 + 16 \cdot 10 + (-7.1) \cdot 4.5}{16 + 52 + 12 + 16 - 7.1} = 6.6$$

Note that the area of the circular void is $\pi \cdot 1.5^2 \simeq 7.1$. This is accounted for as a negative mass.



In the y direction:

$$\frac{16 \cdot 13 + 52 \cdot 7.5 + 12 \cdot 7.0 + 16 \cdot 2 + (-7.1) \cdot 7.5}{16 + 52 + 12 + 16 - 7.1} = 7.4$$

What is the center of gravity?

The *center of gravity* is the point through which the force of gravity acts on an object or system. In most mechanics problems the gravitational field is assumed to be uniform. The center of gravity is then in exactly the same position as the center of mass. The terms center of gravity and center of mass tend to often be used interchangeably since they are often at the same location.

WHAT ABOUT DETERMINING THE CENTER OF MASS FOR A REAL OBJECT?

The *plumb line* method (Figure 5) is also useful for objects which can be suspended freely about a point of rotation. An irregularly shaped piece of cardboard suspended on a pin-board is a good example of this. The cardboard pivots freely around the pin under gravity and reaches a stable point. A plumb line is hung from the pin and used to mark a line on the object. The pin is moved to another location and the procedure repeated. The center of mass then lies beneath the intersection point of the two lines.

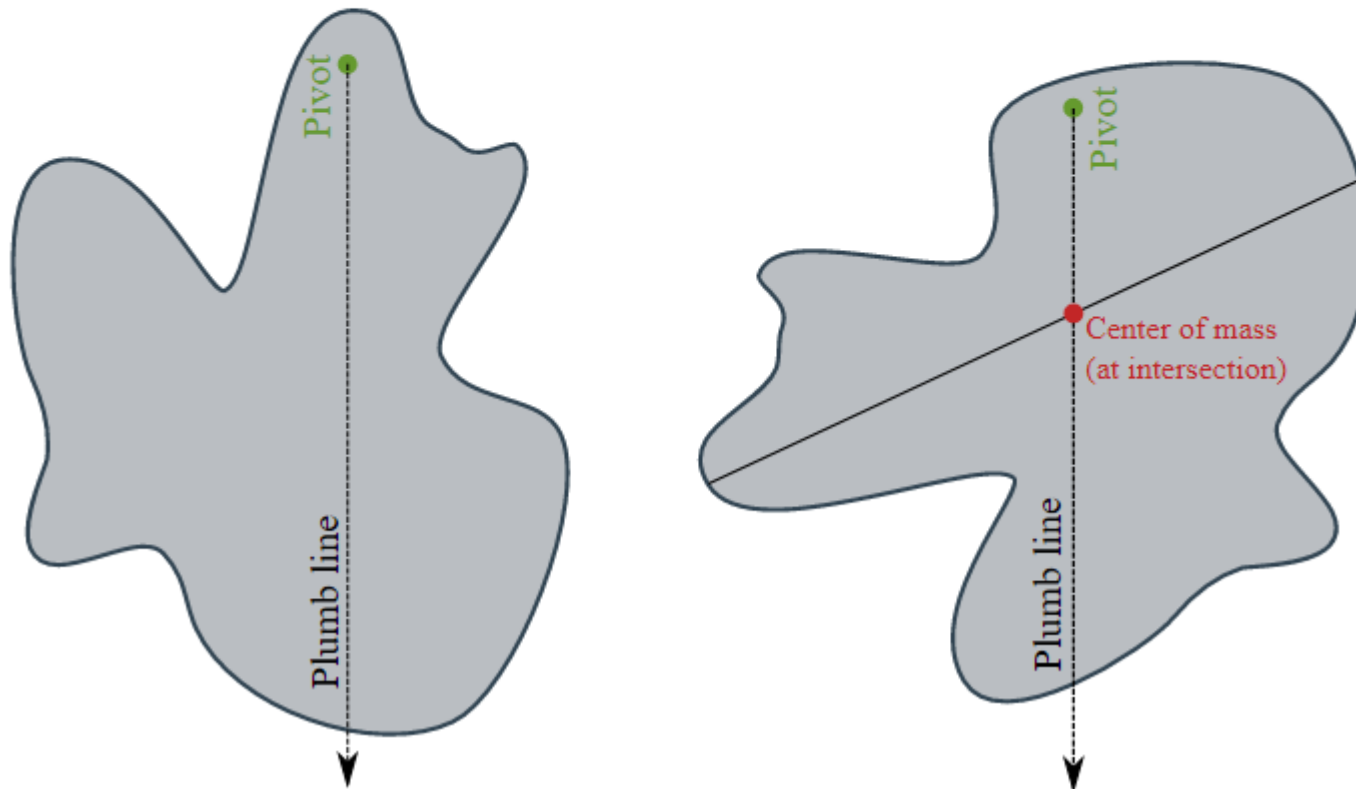


Figure 5: Plumb line method being used to find the center of mass of an irregular object.

Center of mass and toppling stability

One useful application of the center of mass is determining the maximum angle that an object can be tilted before it will topple over.

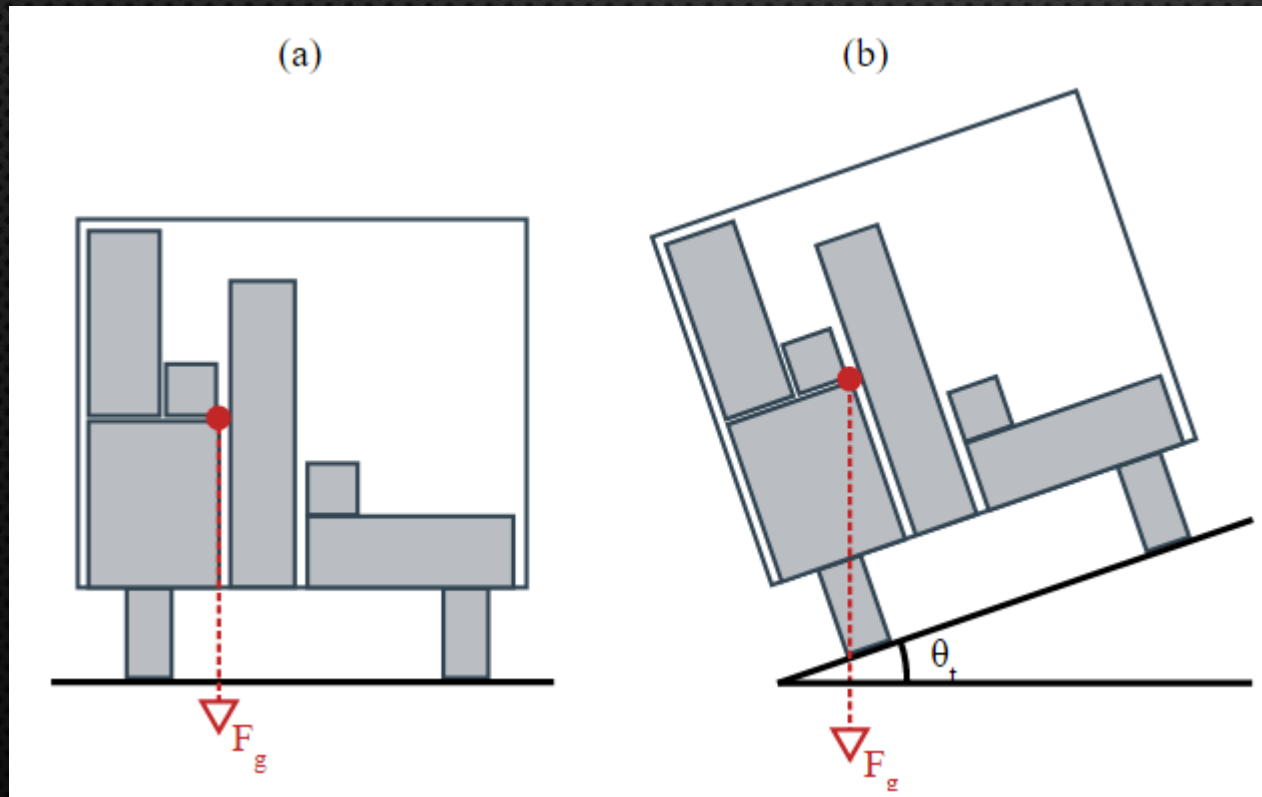


Figure 6: Topple limit of a poorly loaded truck.

Center of mass and toppling stability

Figure 6a shows a cross section of a truck. The truck has been poorly loaded with many heavy items loaded on the left-hand side. The center of mass is shown as a red dot. A red line extends down from the center of mass, representing the force of gravity. Gravity acts on all the weight of the truck through this line.

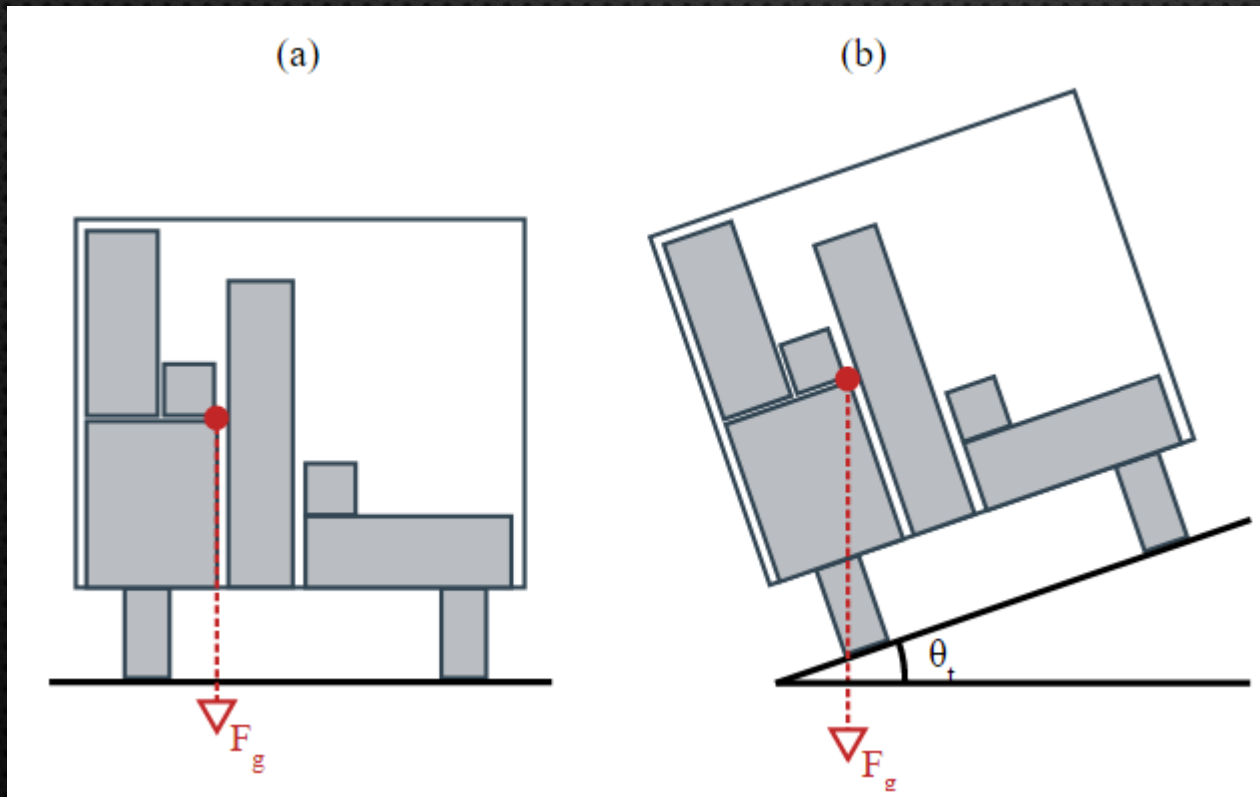


Figure 6: Topple limit of a poorly loaded truck.

Center of mass and toppling stability

If the truck is tipped at an angle θ_t (as shown in figure 6b) then all the weight of the truck will be supported by the left-most edge of the left wheel. Should the angle be further increased then the point of support will move outside of any point of contact with the road and the truck is guaranteed to topple over. The angle θ_t is the *topple limit*.

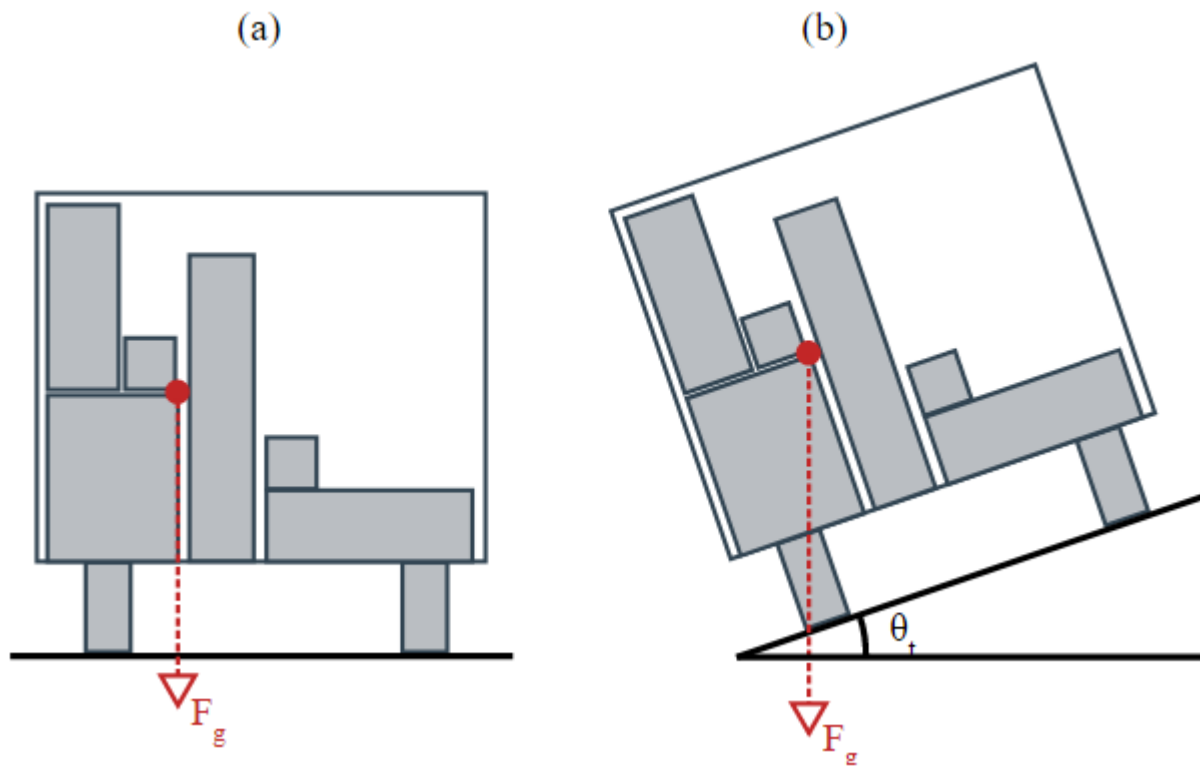
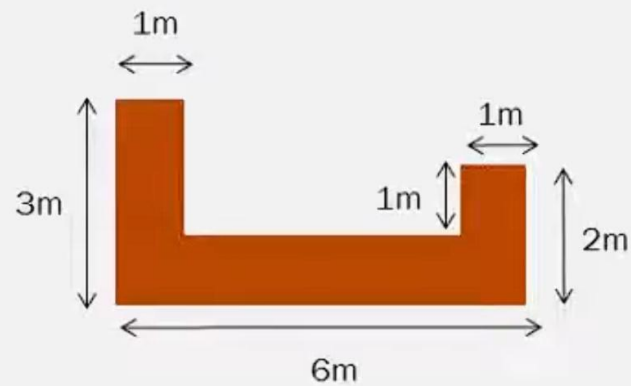


Figure 6: Topple limit of a poorly loaded truck.

Le centre de masse (exemple)

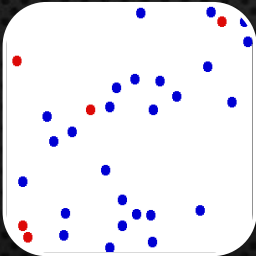
Exemple 10.1 – Centre de masse

Trouvez le centre de masse de la figure homogène suivante.

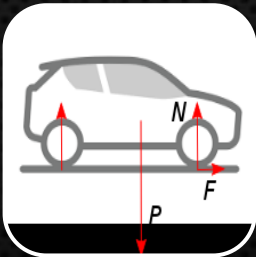


INTERNAL FORCES & EXTERNAL FORCES

Forces



internal: result from interactions between different parts of the same system.



External: result from interactions between the system or part of the system and the environment (outside world).

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Consider two blocks A and B connected together by means of an elastic spring. The whole system is connected to the ceiling by means of a string (Figure 9.16) Determine the external forces acting on the following systems:

- a- Block A.
- b- Block B.
- c- Block A and the spring.
- d- Block B and the spring.
- e- Block A, block B, and the spring.

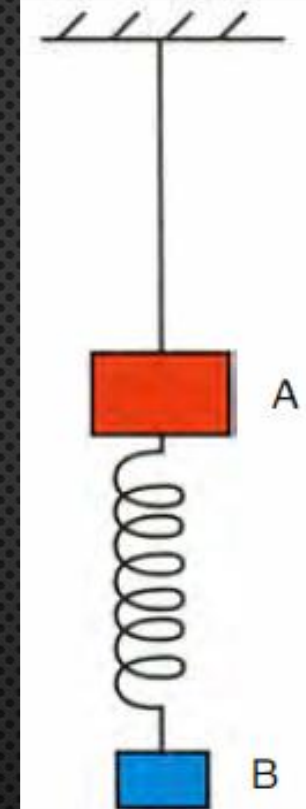


Figure 9.16

CENTER OF GRAVITY

- THE POINT OF APPLICATION OF THE WEIGHT OF A SYSTEM IS CALLED THE **CENTER OF GRAVITY** OF THE SYSTEM.
- FREQUENTLY THE CENTER OF GRAVITY COINCIDES WITH THE CENTER OF MASS OF THE SYSTEM.

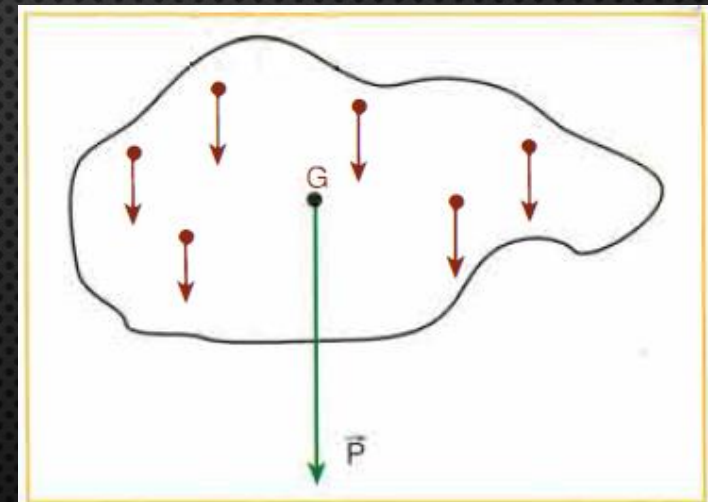


Figure 9.8

The case when the center of gravity coincides with the center of mass.

MOTION OF CENTER OF MASS - THEOREM OF CENTER OF MASS

TO UNDERSTAND ANALYTICALLY THE MOVEMENT OF THE CENTER OF MASS LET US TAKE AGAIN THE DEFINITION OF THE POSITION VECTOR OF THE CENTER OF MASS:

$$\vec{r}_G = \frac{m_1 \cdot \vec{r}_1 + m_2 \cdot \vec{r}_2 + m_3 \cdot \vec{r}_3 + \cdots + m_n \cdot \vec{r}_n}{m_1 + m_2 + m_3 + \cdots + m_n}$$

OR

$$(m_1 + m_2 + m_3 + \cdots + m_n) \cdot \vec{r}_G = m_1 \cdot \vec{r}_1 + m_2 \cdot \vec{r}_2 + m_3 \cdot \vec{r}_3 + \cdots + m_n \cdot \vec{r}_n$$

LET'S DERIVE RELATIVE TO TIME

$$(m_1 + m_2 + m_3 + \cdots + m_n) \cdot \frac{d\vec{r}_G}{dt} = m_1 \cdot \frac{d\vec{r}_1}{dt} + m_2 \cdot \frac{d\vec{r}_2}{dt} + m_3 \cdot \frac{d\vec{r}_3}{dt} + \cdots + m_n \cdot \frac{d\vec{r}_n}{dt}$$

$$\Rightarrow M \cdot \vec{V}_G = m_1 \cdot \vec{V}_1 + m_2 \cdot \vec{V}_2 + m_3 \cdot \vec{V}_3 + \cdots + m_n \cdot \vec{V}_n$$

MOTION OF CENTER OF MASS - THEOREM OF CENTER OF MASS

$$M \cdot \vec{V}_G = m_1 \cdot \vec{V}_1 + m_2 \cdot \vec{V}_2 + m_3 \cdot \vec{V}_3 + \cdots + m_n \cdot \vec{V}_n$$

LET'S DERIVE AGAIN RELATIVE TO TIME, WE GET THEN:

$$M \cdot \frac{d\vec{V}_G}{dt} = m_1 \cdot \frac{d\vec{V}_1}{dt} + m_2 \cdot \frac{d\vec{V}_2}{dt} + m_3 \cdot \frac{d\vec{V}_3}{dt} + \cdots + m_n \cdot \frac{d\vec{V}_n}{dt}$$

$$\Rightarrow M \cdot \vec{a}_G = \underbrace{m_1 \cdot \vec{a}_1}_{\vec{f}_1} + \underbrace{m_2 \cdot \vec{a}_2}_{\vec{f}_2} + \underbrace{m_3 \cdot \vec{a}_3}_{\vec{f}_3} + \cdots + \underbrace{m_n \cdot \vec{a}_n}_{\vec{f}_n}$$

$$\Rightarrow \boxed{M \cdot \vec{a}_G = \sum \vec{F}_{\text{external/System}}}$$

THIS RELATION EXPRESSES THE THEOREM OF THE CENTER OF MASS OF A MATERIAL SYSTEM

MOTION OF CENTER OF MASS - THEOREM OF CENTER OF MASS

$$M \cdot \vec{a}_G = \sum \vec{F}_{\text{external/System}}$$

THEOREM OF CENTER OF MASS:

THE MOVEMENT OF THE CENTER OF MASS OF A SYSTEM IS THAT OF A PARTICLE, OF MASS EQUAL TO THAT OF THE SYSTEM, AND SUBJECTED TO A FORCE EQUAL TO THE SUM OF THE EXTERNAL FORCES APPLIED TO THE SYSTEM.

Example 9.3

A box, of mass $m = 10 \text{ kg}$, is moving along a horizontal track under the action of a horizontal force of traction $T = 20 \text{ N}$. The frictional force with the track is applied at the lower edge of the box and of magnitude $f = 8 \text{ N}$ (Figure 9.11). Determine the acceleration of the center of mass of the box.

0.8 m/s^2

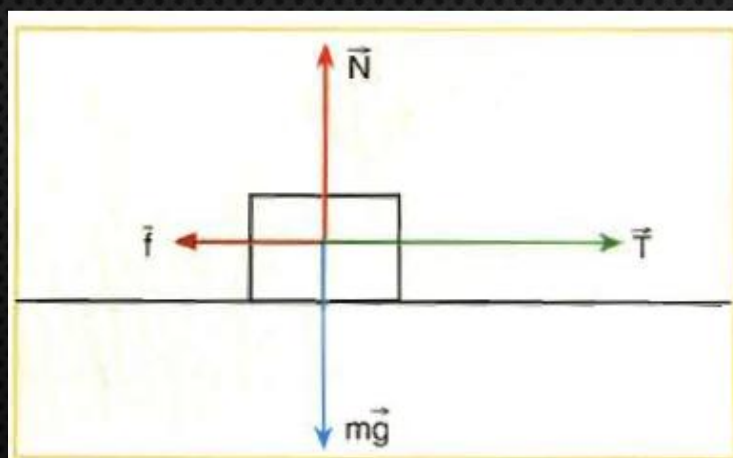


Figure 9.11

Example 9.4

Two particles of masses m_1 and m_2 , move on a horizontal Ox axis under the action of two forces $\vec{f}_1$, applied on m_1 , and $\vec{f}_2$, applied on m_2 (Figure 9.12). Determine the acceleration of the center of mass of the system of the two particles.

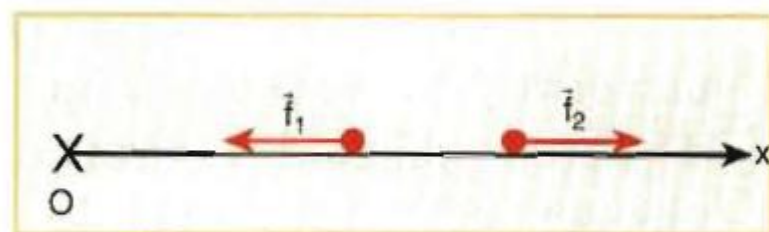


Figure 9.12

MOTION OF CENTER OF MASS - THEOREM OF CENTER OF MASS

$$M \cdot \vec{a}_G = \sum \vec{F}_{\text{external/System}}$$

PARTICULAR CASE

IF THE SUM OF THE EXTERNAL FORCES IS ZERO ($\sum \vec{F}_{\text{ext}} = \vec{0}$), THE CENTER OF MASS OF THE SYSTEM CAN BE EITHER AT REST OR IN UNIFORM RECTILINEAR MOTION. INDEED :

$$\text{If } M \cdot \vec{a}_G = \vec{0} \Rightarrow \vec{a}_G = \vec{0} \Rightarrow \vec{V}_G = \overline{\text{constant}}$$

$$\Rightarrow \left\{ \begin{array}{l} \text{"G" remains at rest if it is initially at rest} \\ \text{or} \\ \text{"G" continues in URM if it is initially moving} \end{array} \right.$$

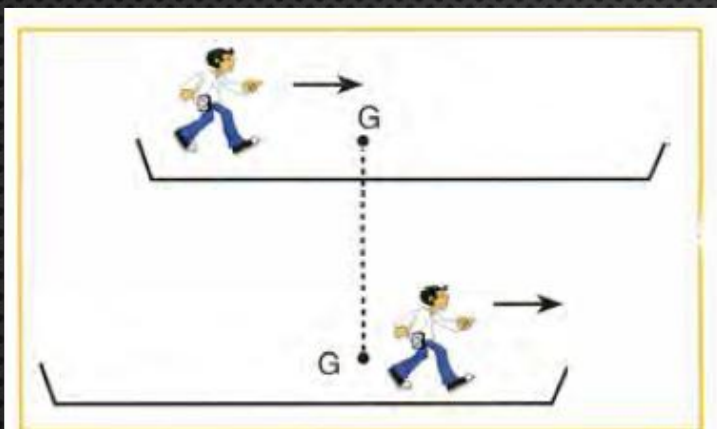


Figure 9.14
The center of mass stays at rest.

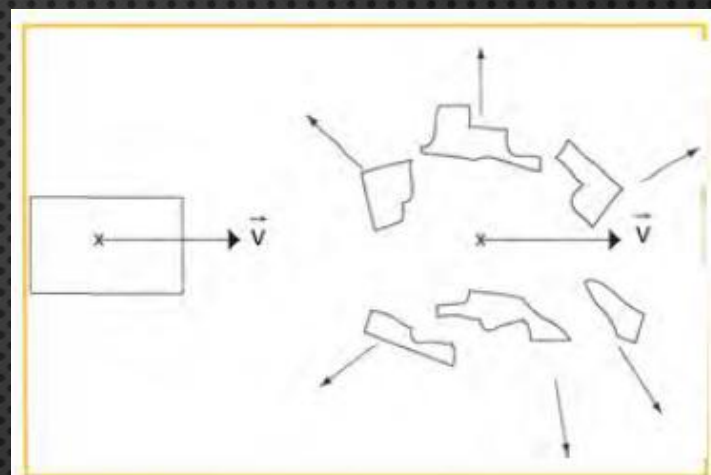


Figure 9.15 - The center of mass continues with a U.R.M.

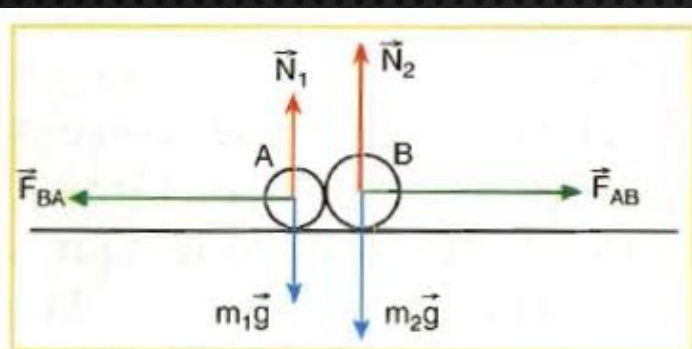


Figure 9.9



Figure 9.13 $\Sigma \vec{F}_{\text{ext}} = \vec{0}$.

17 A train, made up of a locomotive and two cars of mass M each, moves on two horizontal rectilinear rails, under the action of force of traction $\vec{F}$ developed by the locomotive (Figure 9.23).

- a- Determine, in terms of M and F the acceleration of the train and the tension in each of the two connecting cables.
- b- In fact, a constant force of friction $\vec{f}$ exists between each car and the rails. Find the new expressions of preceding quantities (acceleration and tension).

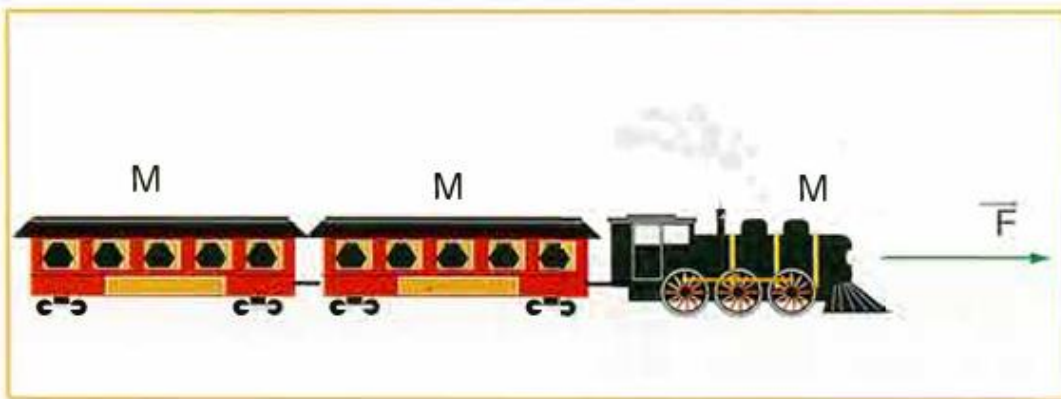


Figure 9.23

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Two particles M and N of respective masses $m_1 = 1$ kg and $m_2 = 2$ kg, move on an axis $X'OX$. In S. I. units, the particle M is affected by two forces $\vec{F}_1 = -2\vec{i}$ and $\vec{F}_2 = \vec{i}$, while the particle N is affected by another two forces $\vec{F}_3 = 6\vec{i}$ and $\vec{F}_4 = -4\vec{i}$.

At $t = 0$ s, the two particles were at rest, M was at the origin O, and N at the point of abscissa $+ 3$ m.

- a- Determine the position of the center of mass of the two particles at $t = 0$ s.
- b- Determine the time equation of motion of each particle separately. i. e x_M and x_N as a function of time t .
- c- Deduce the position of the center of mass at any time t .
- d- Verify the relation: $\sum \vec{F}_{\text{ext.}} = M\vec{a}_G$.